Scientific notation

1 Express the following numbers as a basic numeral:

a 2.49×10^{0} **b** 2×10^{7} **c** 4.13×10^{4} **d** 9×10^{-3}

e 8.97×10^{5} **f** 5.03×10^{4} **g** 3.014×10^{3} **h** 8×10^{6}

i 1.3008×10^{7} **j** 8.36×10^{-1} **k** 7.91×10^{6} **l** 7.14×10^{-3}

2 Complete the following equations:

a $300 = \square \times 10^{2}$ **b** $0.005 = \square \times 10^{-3}$

c $\dfrac{4}{10^{5}} = 4 \times 10^{\square}$ **d** $1 \times 10^{-2} \times 2 \times 10^{-5} = 2 \times \square$

3 Express the following numbers in scientific notation:

a 3×100 **b** $4.5 \times 10\,000$ **c** $7 \div 100$ **d** $6.23 \div 10^{3}$

e 2000 **f** 0.002 **g** $\dfrac{1}{10\,000}$ **h** $884\,000$

i $84\,626\,000$ **j** 6.14 **k** 0.000347 **l** $\dfrac{3}{100\,000}$

m 0.07 **n** 0.764 **o** $2\,000\,000$ **p** $\dfrac{1}{10\,000}$

4 10^{5} is equal to:

A 1 million **B** 10 thousand **C** 10 million **D** 100 thousand

5

a Determine whether the following numbers are equivalent to 0.00045:

i 4.5×10^{-4} **ii** 45×10^{-4} **iii** 4.5×10^{-6} **iv** 0.45×10^{-3}

v 4.5×10^{-3} **vi** 45×10^{-5}

b Of the expressions selected in part (a), which one is in scientific notation?

6 For each of the following pair of numbers:

i State the largest number. **ii** Find the factor by which it is greater.

a 780×10^{-1} and 7.8×10^{2} **b** 4×10^{-3} and 4×10^{-7}

c 3×10^{5} and 9×10^{9}

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7 For each of the following set of numbers:

i State the largest number.

ii State the smallest number.

a

- 7.27×10^{-7}
- 2.22×10^{-7}
- 4.76×10^{-7}

b

- 1.25×10^{-2}
- 1.25×10^{-6}
- 1.25×10^{2}

c

- 9.37×10^{-4}
- 6×10^{-4}
- 6×10^{-9}

d

- 7.31×10^{-9}
- 5.13×10^{-9}
- 8.31×10^{-9}

e

- 2.93×10^{-2}
- 7.88×10^{2}
- 7.88×10^{-2}

f

- 8.99×10^{-3}
- 1.99×10^{-6}
- 8.99×10^{-6}

g

- $4.7 \times 10^{2} \times 10^{3}$
- $4.7 \times 10^{-8} \times 10^{8}$
- $4.7 \times 10^{9} \times 10^{8}$

h

- $2.12 \times 10^{-1} \times 10^{7}$
- $2.12 \times 10^{-7} \times 10^{4}$
- $2.12 \times 10^{7} \times 10^{2}$

8 Evaluate the following, giving your answers in scientific notation:

a $2 \times 10^{6} \times 3 \times 10^{5}$

b $4 \times 10^{-3} \times 4 \times 10^{-5}$

c $\dfrac{15 \times 10^{6}}{5 \times 10^{3}}$

d $\dfrac{9 \times 10^{-3}}{3 \times 10^{-5}}$

e $\left(9 \times 10^{9}\right) \times \left(5 \times 10^{-6}\right)$

f $\left(7 \times 10^{5}\right) \times \left(3 \times 10^{4}\right)$

g $\dfrac{800\,000 \times 180\,000\,000}{6\,000\,000\,000}$

h $11\,000\,000 \times 0.004 \times 0.0005 \times 20\,000$

Scientific notation and calculators

9 A calculator dislays scientific notation in the form aEb, express the following numbers as a basic numeral:

a $5.81E5$ **b** $3.16E\left(-4\right)$ **c** $2.5E6$ **d** $7.05E\left(-3\right)$

10 Use your calculator to find the value of the following:

a $\left(6.20 \times 10^{-2}\right)^{2}$

b $\sqrt[3]{2.197 \times 10^{-6}}$

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11 Use your calculator to evaluate each of the following. Express your answer in scientific notation, rounded to four significant figures.

a $0.00051^2 \times 0.0081^4$

b $\dfrac{6.45 \times 10^9 \times 4.3 \times 10^{-3}}{8.3 \times 10^4}$

12 A calculator displays $3.35E5$ as the correct final answer when solving the following problem. Find the missing values in the problem to produce the given answer.
"At any particular time, approximately $2.32 \times 10^{\blacksquare}$ aircraft are registered as being in flight over $\blacksquare \times 10^{10}$ square metres of air space. On average, how much air space does each aircraft have?"

Significant figures

13 State the number of significant figures displayed in each of the following values:

a 4.25 **b** 84.00 **c** 0.002556 **d** 0.0702

14 Round the following values to the specified number of significant figures:

a 4.95 to two significant figures

b 0.006037736 to two significant figures

c $801\,600\,713$ to four significant figures

d 0.050728 to three significant figures

15 For each of the following expressions:

i State the number of significant figures given in the least precise measurement.

ii Hence, calculate the result, rounded to the least number of significant figures.

a $9.17\text{ m} \times 9.790\text{ m} \times 4.70\text{ m}$

b $418\text{ mL} \times 5.7 \div 934.5$

16 Evaluate each of the following. Express your answer in scientific notation, rounded to three significant figures.

a 215^4

b 15^{-4}

17 A number has been rounded to two significant figures. If the rounded number is 4500, find the largest possible value of the original number.

18 A number, when rounded to one significant figure, is 500. The original number is a multiple of 4, and halving it gives you a square number. Find the original number.

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19 A number, when rounded to one significant figure, is 3000.

a Find the largest possible value of the original number.

b Find the smallest possible value of the original number.

c Find the difference between the largest and smallest values.

20 Consider the equation $9^n = 11$.

a State the closest whole number to n.

b The value of n is given by $n = \dfrac{\log 11}{\log 9}$. Using the log button on your calculator, find the value of n. Round your answer to three significant figures.

Applications

21 A number is represented in the form $2E\,(-6)$. Which of the following could this number represent?

- The mass of an ant in grams
- Earth's population
- The distance in kilometres between two planets
- The diameter of a human cell

22 The world's oceans hold approximately $1\,332\,000\,000$ cubic kilometres of water.

a Express this volume of water in scientific notation.

b Given $1\ \text{km}^3 = 1\,000\,000\,000\,000\ \text{L}$, how many litres of water do the world's oceans hold? Express your answer in scientific notation.

23 Bone marrow in a person's body produces 2.3×10^6 red blood cells each second.

a Which of the following would be the closest rounded value for this number?

- 20 million each second
- 2 million each second
- 2 hundred thousand each second

b Hence, estimate the number of red blood cells produced each minute.

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24 The tiny tubes in our kidneys measure about 1.2×10^{-5} metres in diameter.

a Which of the following would be the closest rounded value for this number?

- 1 ten-thousandth of a metre
- 1 millionth of a metre
- 1 hundred thousandth of a metre

b Hence, estimate the diameter of the tubes in millimetres.

25 If we round to one significant figure, sound travels at a speed of approximately 0.3 kilometres per second, while light travels at a speed of approximately $300\,000$ kilometres per second.

a Express the speed of sound in kilometres per second in scientific notation.

b Express the speed of light in kilometres per second in scientific notation.

c How many times faster does light travel than sound?

26 A country's land size is approximately $250\,000$ square kilometres and its population is approximately 68 million people.

a Express the number of square kilometres in scientific notation.

b Express the population in scientific notation.

c Hence, determine the population density measured in persons per square kilometre. Express your answer in scientific notation.

27 Consider the following:

- A plane is travelling eight hundred thousand metres per hour.
- An asteroid travelling at 8×10^8 metres per hour.

a Express the speed of the plane in scientific notation.

b How many times faster is the asteroid travelling? Express your answer as a basic numeral.

28 Two stars, A and B, in neighbouring galaxies have masses of 2.7×10^{41} kg and 8.91×10^{44} kg respectively. How many times more massive is star B than star A? Express your answer as a basic numeral.

29 The mass of the largest mammal on Earth is approximately 1.5×10^3 times greater than the mass of an average adult human who weighs 90 kg. According to this information, find the approximate mass of the largest mammal on Earth. Express your answer as a basic numeral.

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30 An insect has an average mass of 0.2 gram. During a plague, it was estimated that there were 100 million insects in a particular area. Find the total mass of insects in scientific notation.

31 An estimated 9.1×10^7 people watched the final of the last World Cup, while an estimated 6.8×10^7 people watched the final of the World Cup four years before. Find the average number of people who watched the final of the World Cup the last two times it occurred.

32 Last year, the total CO_2 emissions of a country was approximately 7×10^{11} kg and the country had approximately 3.5×10^8 citizens. Find the average CO_2 emissions for each citizen of the country.

33 The mass of an atom of helium is 6.6465×10^{-24} grams and each litre of helium weighs 145 grams. A balloon is pumped with 11 litres of helium.

- **a** Find the number of grams of helium in the balloon.
- **b** Find the number of atoms in the balloon. Express your answer in scientific notation, rounded to one significant figure.

34 One mole (mol) of an element consists of approximately 6.02×10^{23} atoms.

- **a** Find the number of atoms in 3.5 mol of element X. Express your answer in scientific notation.
- **b** If 3.5 mol of element X weighs 31.5 g, find the mass of 1 mol of element X.
- **c** Find the approximate mass of one atom of element X. Express your answer in scientific notation, rounded to four significant figures.

35 The Earth orbits the Sun at an approximate speed of 8333 metres per second.

- **a** Find how far it travels in one hour. Leave your answer as a basic numeral.
- **b** Find how far it travels in 23 hours. Express your answer in scientific notation, rounded to one significant figure.

36 Asteroids A and B are currently 4×10^{14} km apart. If they continue in a straight line towards one another, travelling at 3857 km/h and 7938 km/h respectively, how many days will it take for them to collide? Express your answer in scientific notation, rounded to five significant figures.

37 Planet A has a radius of 8.2×10^{18} km, while planet B has a radius of 4.1×10^{23} km. How many times greater than the volume of planet A is the volume of planet B? Express your answer in scientific notation. The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$.

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